Vinayak Dhruv

LinkedIn: vinayak-dhruv-a257bb174 • Email: vkdhruv101@gmail.com • Mobile: +91 9319567708

Education

Delhi Technological University Delhi, India
B.Tech in Electrical Engineering *2018 - 2022*

- **GPA**: 8.08/10.00
- **Highlighted Coursework**: Mathematics - 1 & 2, Programming Fundamentals, Probability Theory, Basic Statistics, Numerical Optimization Methods, Machine Learning, Operative Systems, Introduction to Deep Learning, Bayesian Methods of Deep Learning, Software Engineering, Swarm & Evolutionary Computing, Big Data Analytics, Computer Vision, Logical Reasoning
- **Research**: Autonomous navigation systems for single lanes, COVID antiviral drug dataset and prediction system, Traffic assistance system for managing toll tax, LLMs, Generative AI, IoT Systems

Arwachin International School Delhi, India
Sr. Secondary *2017 - 2018*

- **Percentage**: 92.4%
- **Highlighted Coursework**: Physics, Chemistry, Mathematics, Computer Science, Physical Education, English

Experiences

Work Experience:

OLA Krutrim Bangalore, India
AI Research Scientist *Jun 2022 – Present*

- Working as a Research Scientist at the AI division of OLA for developing foundational LLM models, responsible for multiple research projects in NLP as well as Speech in Generative AI, like TTS systems, Chat Bots, etc. As part of a research team of 5 built India's first LLM based Chatbot for indic languages.[Link].
- As part of the AI division developed a project for live-streaming lip synced videos for multiple languages for a single speaker. Worked to improve Wav2Lip on better resolution videos by 12%. Sample posted online [Link].
- Led the efforts on building a LLM based solution for Customer Care services at OLA. Build and innovated ideas that resulted in automation of 60% of customer interactions.
- Headed efforts on improving Driver Allocation model for ride allocations, and worked on building better features and boosted the XGboost scores, which brought a 5% increase in the revenue.
- Researched and devised an autonomous Proximity Unlock solution for OLA e-Scooters, with an accuracy of 93.7%. [Link]
- Worked and supervised on Criticality Score prediction for emails and customer queries , assisted in building a response system and auto-reply pipeline for emails, with a 90+% accuracy in their criticality categorization.
- Built projects regarding SoH prediction of batteries, time to reach 100% charge and Health analysis with 90+% accuracy and helped in deploying them in production.
- Spearheaded the VoiceBot project for customer care replying and answering purposes utilizing whisper and GPT-3, and worked on transcribing and developing a database for the project using pre recorded audios at the company.

Amazon Ads Bangalore, India
ML Engineer Intern *Jan 2022 – Jun 2022*

- Worked towards forecasting Out of balance dates for the advertisers at Amazon, using time series prediction to predict their spends, with an accuracy of 92.4% and pushed it to production over cloud infrastructure.
- Built a pipeline on AWS applications and compared performance of the model on various metrics.

VenueMonk Delhi, India
Data Science Intern *Dec 2020 – Mar 2021*

- Automated 100% of the image handling and validation system of the company for their website [Link] utilizing image processing techniques.
- Built a venue ranking system for the company, and designed a pipeline to handle the ranking system for different locations in India.
- Made clustering models for venues with a score of 92% accuracy and deployed various models on the cloud using AWS, building recommendation systems for users on the website.

The TravelBack Vancouver, Canada
Data Science Engineer Intern *Oct 2020 – Dec 2020*

- Led the data mining team and developed pipelines for data mining from various restaurant, hotel and hiking related websites.
- Developed algorithms to map hotels and hiking locations in the Vancouver region, using recommendation systems for customers

Research Work:

Unmanned Ground Vehicle Team DTU, India
Team Captain *Sept 2018 – April 2022*
Faculty Advisor: Professor S. Indu

- Worked as the Team Captain in developing Unmanned vehicles and Autonomous systems on ROS [Link]], designing algorithms and building navigation stacks for obstacle avoidance and path planning. Designed models for lane reconstruction and a self-built monocular depth estimation model to help in autonomous navigation.
- Led a team of 25 students on IGVC-2020 Oakland, and also won the 9th prize overall. Part of team in 2019 IGVC built a semi-autonomous bot for handling lane following tasks and obstacle avoidance.[Link]
- Developed and worked on a WeedBot for classifying weeds in the farms and worked on developing a system to scan the area and detect the regions infested.
- Participated in Micromouse and similar national competitions on robotics and simulations on problems like maze solving and path finding, as team lead of UGV-DTU [Link].

XERO Technologies Delhi, India
ML Engineer *Nov 2020 – Mar 2021*
Start-up XERO Tech

- Developed a Startup idea around Traffic Count and Classification Application [Link], an application that supports the tracking, classification, and counting of vehicle classes on visual data derived from Indian Roadways.
- Worked on Image analysis and night based image detection by image enhancement and data augmentation.
- Led the Vision team of the company for developing Object detection for traffic vehicles and developed an algorithm to keep track of the detected vehicle and also determine the total toll collected.

CALIBRE Research Community DTU, India
Research Assistance *Jan 2019 – June 2019*
Faculty Advisor: Dr. Rahul Katarya

- Researched and assisted in NLP related projects like sentiment analysis and comparative study of ML algorithms to solve it, developed my own model for analysing Meta tweets where I got 92.3% acc.
- Developed a COVID-antiviral drug prediction model with my partner in research, curated a complete dataset from various sites and open source datasets, built a Bi-LSTM based classification model to predict the antiviral drug set best suited for any genome chain with 94.5% acc.

Projects

A Robust Underwater Plastic Detection System
to better distinguish between Jellyfish and plastic bags for underwater detection for trash [Link]
Prof. Biplab Banerjee, IIT Bombay

- Worked and initiated a project for developing robust solutions for underwater organism detection.
- Analysed and found that 90% of available datasets are biased and do not consider negative classes properly. Developed a better dataset for countering the plastic bag classification problem.

Loan Amount Sanction System
to predict the loan amount for a loan applicant [Link]

- Developed a Loan amount sanctioning predictor based on ML methods like XGBoost for loan amount sanctioned for houses considering various features like Credit Score, Income, etc using Exploratory Data Analysis.
- The model predictions gave a score of 87% on CIPLA Data Scientist Challenge.

Smart Face Lock
for preventing privacy attacks on phones [Link]

- Attempted to detect live faces on phones, using a KNN model for recognition of familiar faces.
- A simple CNN model trained on a dataset of Live images and Fake images(like photographs) is used on the recognized images to classify if it's live, with a 0.2% error rate.

Soldier Strap
finalist for Smart India Hackathon 2020 [Link]

- Worked as a team to develop a solution for soldier straps that could help maintain optimum patterns for walking, and keep track of weak points using clustering algorithms.
- Won the college finalist position from DTU for SIH 2020 for the idea and prototype results.

Centaur - Unmanned Rover
an unmanned ground vehicle funder Prof. S.Indu for IGVC 2020 [Link]

- Engineered an unmanned rover, in collaboration with my team UGV-DTU, for competitive purposes.
- Handled the lane detection and navigation stack on the ROS framework, to travel on grass lanes accurately.

Youtube Video Review Score Generator
an analysis project over Youtube videos [Link]

- Made an analysis model to generate scores for a product based on the reviews by customers on YouTube videos for it.
- Performed a sentiment analysis using a simple LSTM on extracted transcripts of the videos, with 82% acceptance rate.

Publications

- V. Dhruv, R. Awasthi and S. Pawar, **User Intent-Based Proximity Unlock: A Novel Approach For Secure Vehicle Unlocking**, IOSR Journal of Computer Engineering (IOSR-JCE), e-ISSN: 2278-0661,p-ISSN: 2278-8727, Volume 25, Issue 6, Ser. 1 (Nov. – Dec. 2023), PP 32-37 [Link]
- H. Kaushal, V. Dhruv and I. Sreedevi, **Minotaur - A Single Lane Navigation System**, 2022 8th International Conference on Automation, Robotics and Applications (ICARA), 2022, pp. 113-117. [Link]
- V. Dhruv, A. Jha, and D. Chandel, **Traffic Count and Classification Application**, Technical Report, Xero Technologies, 2021. [Link]

Awards, Honors, and Scholarships

- Gold Star at Hackerrank Problem solving in C++ and 3 star at CodeChef with best Global Rank as 354
- As a member of YETi was a College Finalist at Smart India Hackathon 2020.
- As captain of UGV won 9th position overall with 5th in Cyber Security in IGVC-2019 at Michigan, Oakland University.
- Won the first place in NASA Astronomy Olympiad, at Senior Secondary school.

Extracurricular Activities

- As the team captain led my volleyball team into Zonals — Delhi 2016
- Participated and won Bronze in Relay race at school — 2015
- Badminton Tournament runner up from my House — 2016
- Maths Olympiad winner — 2017 and 2016
- National Astronomy Olympiad, Gold winner — 2017 and 2016

Positions of Responsibility

- Team Captain of Team UGV-DTU from 2020 to 2022 (continued as team advisor) and led the team into many competitions
- Class representative for 5th semester CO327 Course — Aug 2020 - Dec 2020
- Team Captain for my Volleyball team in my 10th standard — March 2015 - March 2016
- Head Boy of my school — academic year 2017-2018
- Head of Disciplinary Committee — academic year 2016-2017

Community Service

- Worked as Fundraiser on multiple occasions for literary fests and technical gatherings in DTU — 2018 - 2019
- Tutored kids who could not afford education for free around my residence — Summer of 2014
- Worked as a member of the Eco Club and planted saplings across the residence spreading awareness — 2017

Skills and Interests

Computer Vision, Prompt Engineering, Generative AI, Deep Learning, Machine Learning, Data Structures and Algorithms, LLM Basics, AWS, ROS, Python, Pytorch, R, SQL, C++, Active Listening.